

High order derivatives (13)

The derivative of a function f leads to another function f' .

If f' has a derivative, it is denoted by f'' and is called the Second derivative of f . Thus

$$f''(x) = D_x(f'(x)) = D_x(D_x f(x))$$

For every x such that the indicated derivatives exist.

The derivative on the right in the above equation is abbreviated $D_x^2 f(x)$.

In the like manner the third derivative f''' of f is the derivative of the second derivative.

i.e.

$$f'''(x) = D_x(f''(x)) = D_x(D_x^2 f(x)) = D_x^3 f(x).$$

In general, if n is a positive integer, then $f^{(n)}$ denotes the n^{th} derivative of f and is found by starting with f and differentiating, successively, n times.

$$\text{i.e. } f^{(n)}(x) = D_x^n f(x).$$

The integer n is called the order of the derivative $f^{(n)}(x)$.

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Thus, if $y = f(x)$, then the first n derivatives are denoted by

$$D_x y, D_x^2 y, D_x^3 y, \dots, D_x^n y$$

$$\text{or } y', y'', y''', \dots, y^{(n)}$$

If the differential notation is used we write

$$\frac{dy}{dx}, \frac{d^2 y}{dx^2}, \frac{d^3 y}{dx^3}, \dots, \frac{d^n y}{dx^n}.$$

NB: In this case, the symbols used for higher derivatives should not be interpreted as quotients.

Example:

1. If $f(x) = 4x^2 - 5x + 8 - (3/x)$, find the first four derivatives of $f(x)$.

Solution

$$\text{Since } f(x) = 4x^2 - 5x + 8 - 3x^{-1}.$$

$$\Rightarrow f'(x) = 8x - 5 + 3x^{-2} = 8x - 5 + \frac{3}{x^2}.$$

$$f''(x) = 8 - 6x^{-3} = 8 - \frac{6}{x^3}.$$

$$f'''(x) = 18x^{-4} = \frac{18}{x^4}$$

$$f^{(4)}(x) = -72x^{-5} = -\frac{72}{x^5}.$$

2. Find the first eight derivatives of $f(x) = \sin x$.

Solution

$$f(x) = \sin x$$

$$f'(x) = \cos x$$

$$f''(x) = -\sin x$$

$$f'''(x) = -\cos x$$

$$f^{(4)}(x) = -(-\sin x) = \sin x$$

$$f^{(5)}(x) = \cos x$$

$$f^{(6)}(x) = -\sin x$$

$$f^{(7)}(x) = -\cos x$$

$$f^{(8)}(x) = -(-\sin x) = \sin x$$

Exercise:

1. Find $D_x y$, $D^2 x y$, $D^3 x y$ for each of the functions

(a) $y = 2x^5 + 3x^3 - 4x + 1$

(b) $y = \sqrt{2 - 5x}$

(c) $y = \cos 3x$

(d) $y = \sin(x^2)$

(e) $y = (2x-3)/(3x+1)$

(f) $y = 1/(x^2+4)$

(g) $y = \sqrt[3]{2-9x}$

(h) $y = (3x+1)^4$

Implicit Differentiation

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Given an equation $y = 2x^2 - 3$

it is customary to say that y is a function of x .

i.e. $y = f(x) = 2x^2 - 3$.

The equation $4x^2 - 2y = 6$ defines the same function f

For such a function, we say that y is an explicit function of x .

Not all function can be expressed as an explicit function.

Consider the following function

$$x^2 + y^2 = 4$$

$$\Rightarrow y = \sqrt{4 - x^2} \text{ or } y = -\sqrt{4 - x^2}$$

Each of these functions is an explicit function, but $x^2 + y^2 = 4$ is an implicit function

Consider the equation $4x^2 - 2y = 6$. Find $\frac{dy}{dx}$.

Solution

To differentiate this function, we

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differentiate both sides with respect to x

$$1.e \frac{d}{dx}(4x^2 - 2y) = \frac{d}{dx} 6.$$

$$\Rightarrow 8x - 2 \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = 4x$$

So given $x^2 + y^2 = 4$, we can still find the derivative with respect to x .

$$1.e \frac{d}{dx}(x^2 + y^2) = \frac{d}{dx} 4.$$

$$\Rightarrow 2x + 2y \frac{dy}{dx} = 0$$

$$2y \frac{dy}{dx} = -2x \Rightarrow \frac{dy}{dx} = \frac{-x}{y} : y \neq 0$$

NB: When differentiating y , we must think of y as a function of x .

Examples:

1. Find y' in each of the following
- (a) $x^3 - 2xy + y^3 = 4$.

Solution

Differentiating on both sides w.r.t x .

$$\frac{d}{dx}(x^3 - 2xy + y^3) = \frac{d}{dx} 4.$$

$$3x^2 - 2\left[y + x \frac{dy}{dx}\right] + 3y^2 \frac{dy}{dx} = 0$$

$$3x^2 - 2y - 2x \frac{dy}{dx} + 3y^2 \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} (3y^2 - 2x) = 2y - 3x^2$$

$$\Rightarrow \frac{dy}{dx} = \frac{2y - 3x^2}{3y^2 - 2x} ; 3y^2 - 2x \neq 0$$

b) $y = x \sin y$
Solution

$$\frac{d}{dx} y = \frac{d}{dx} x \sin y$$

$$\Rightarrow \frac{dy}{dx} = \sin y + x \cos y \frac{dy}{dx}$$

$$(1 - x \cos y) \frac{dy}{dx} = \sin y$$

$$\Rightarrow \frac{dy}{dx} = \frac{\sin y}{1 - x \cos y}$$

NB: The derivative of $2xy$ and $x \sin y$ involves the product rule.

c) $4xy^3 - x^2y + x^3 - 5x + 6 = 0$.

Solution

$$\frac{d}{dx} (4xy^3 - x^2y + x^3 - 5x + 6) = 0$$

$$4 \left[y^3 + 3xy^2 \frac{dy}{dx} \right] - \left[2xy + x^2 \frac{dy}{dx} \right] + 3x^2 - 5 = 0.$$

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$$4y^3 + 12xy^2 \frac{dy}{dx} - 2xy - x^2 \frac{dy}{dx} + 3x^2 - 5 = 0$$

$$(12xy^2 - x^2) \frac{dy}{dx} = 2xy - 4y^3 - 3x^2 + 5$$

$$\Rightarrow \frac{dy}{dx} = \frac{2xy - 4y^3 - 3x^2 + 5}{12xy^2 - x^2}$$

d) Find the slope of the tangent line to the graph of $y^4 + 3y - 4x^3 = 5x + 1$ at $p(1, -2)$.

Solution

Remember $\frac{dy}{dx}$ at $p(1, -2)$ = gradient of the tangent line to the curve at $p(1, -2)$.

$\Rightarrow$ We find $\frac{dy}{dx}$ at substitute $x=1$ and $y=-2$.

$$\frac{d}{dx} (y^4 + 3y - 4x^3) = \frac{d}{dx} (5x + 1)$$

$$4y^3 \frac{dy}{dx} + 3 \frac{dy}{dx} - 12x^2 = 5$$

$$(4y^3 + 3) \frac{dy}{dx} = 12x^2 + 5$$

$$\Rightarrow \frac{dy}{dx} = \frac{12x^2 + 5}{4y^3 + 3}$$

$$\Rightarrow \frac{dy}{dx} \bigg|_{x=1, y=-2} = \frac{12(1)^2 + 5}{4(-2)^3 + 3} = \frac{17}{-29}$$

(e) Find y' and y'' if $x^3 - y^3 = 8$.

Solution

Differentiating both sides with respect to x

$$\frac{d}{dx} (x^3 - y^3) = \frac{d}{dx} 8$$

$$\Rightarrow 3x^2 - 3y^2 \frac{dy}{dx} = 0$$

$$\Rightarrow 3x^2 = 3y^2 \frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{x^2}{y^2} : y \neq 0.$$

$$\Rightarrow y' = \frac{x^2}{y^2}.$$

Differentiating again with respect to x .

$$\Rightarrow \frac{d}{dx} y' = \frac{d}{dx} \frac{x^2}{y^2}$$

$$\Rightarrow y'' = \frac{2xy^2 - 2x^2y \frac{dy}{dx}}{(y^2)^2}.$$

$$= \frac{2xy^2 - 2x^2y \times \frac{x^2}{y^2}}{(y^2)^2}$$

$$= \frac{-16x}{y^5} : y \neq 0.$$

(f) Given $\sec(xy) = 3x$, find $\frac{dy}{dx}$.

Solution

Differentiate both sides with respect to x .

$$\frac{d}{dx}(\sec xy) = \frac{d}{dx}(3x)$$

$$\sec(xy) \tan(xy) \left[y + x \frac{dy}{dx} \right] = 3$$

$$y \sec(xy) \tan(xy) + x \sec(xy) \tan(xy) \frac{dy}{dx} = 3$$

$$\frac{dy}{dx} = \frac{3 - y \sec(xy) \tan(xy)}{x \sec(xy) \tan(xy)}$$

Exercise:

1. Use implicit differentiation to find $\frac{dy}{dx}$.

(a) $x^2 y^3 - 2x + y = 3$. Ans $\frac{2(1 - xy^3)}{1 + 3x^2 y^2}$

(b) $y = x^3 - y^3$ Ans $\frac{3x^2}{1 + 3y^2}$

(c) $y = \sqrt{x+y}$ Ans $\frac{1}{2\sqrt{x+y} - 1}$

(d) $x \sin y + y \sin x = 1$. Ans $-\frac{y \cos x + \sin y}{x \cos y + \sin x}$

(e) $y^2 = x \cos y$

(f) $y = e^{xy}$

(g) $y = \ln(xy)$

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2. Find an equation of the tangent line to the graph of the given equation at the indicated point p .

(a) $xy + 16 = 0$, $p(-2, 8)$.

$$\text{Ans} = 4x - y + 16 = 0$$

(b) $2x^3 - x^2y + y^3 - 1 = 0$, $p(2, -3)$.

$$y + 3 = \frac{-36}{23}(x - 2)$$